

Digital Filters

Physical Measuring Techniques FK7063

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June 2026

Introduction

In this computational lab, we construct several digital filters using Matlab and compare them to each other. We also analyze data from an EKG signal and determine ways to remove noise from it.

1 Tasks

1.1 Constructing a Second Order Stop-Band Filter

We define the frequency response of a stop-band filter whose attenuation is 0 at $f = f_0 = 50$ Hz with a bandwidth of $b = 4$ Hz at -3 dB (corresponding to an attenuation of $1/\sqrt{2}$). The sampling rate is $F_s = 1000$ Hz.

In the space of the z-transform, the frequency f_0 corresponds to $\exp(i\Omega_0)$ with $\Omega_0 = 2\pi \cdot f_0/F_s$. To get the desired attenuation, we place a zero at this point, i.e. we add a factor $(1 - \exp(i\Omega_0))$ to the denominator. Since real signals always contain negative frequencies in the same amount as corresponding positive frequencies, we also need to attenuate -50 Hz (corresponding to $-\Omega_0$) in the same way, so we add a factor $(1 - \exp(-i\Omega_0))$.

2 Conclusion